

Online Recruitment Fraud (ORF) Detection Using Transformer Models With SHAP Explainability

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Abstract — With the increase of online job recruiting sites, job seekers have become more exposed to fake job postings that are ultimately leading to financial fraud, identity theft, and abuse of their data. Conventional machine learning strategies that use hand-crafted features and shallow textual representations are unable to reach beyond the surface and extract deeper contextual representation of meaning in job posts. The project applies transformer-based deep learning architecture, namely Bidirectional Encoder Representations from Transformers (BERT) and Efficiently Learning an Encoder that Classifies Token Replacements Accurately (ELECTRA), to detect fake job postings with greater accuracy. A class imbalance was mitigated using Borderline- SMOTE (SMOBD) and model interpretability of the classifiers' prediction was displayed via SHAP explainability. The models were fine-tuned using a benchmark job postings data, and performance was evaluated using precision, recall, F1 score, and accuracy measures. The experimental results indicate superior performance of ELECTRA for fraudulent detections as it was discriminator-based, while BERT was superior regarding contextual semantic understanding of a job descriptions. An analysis of performance comparisons show that transformer-based models yield higher performance recall and balanced accuracy than conventional approaches, allowing for reduced risk of the consequences of false negatives in classification where fraudulent jobs are incorrectly classified as legitimate. This study provides compelling evidence that transformer model architectures are a viable approach to mitigate fake job postings on online job recruitment platforms.

Keywords — BERT, ELECTRA, Online Recruitment Fraud, Transformer models, Deep learning, SHAP Explainability.

I. INTRODUCTION

A. Motivation and Problem Statement

The hiring process has been altered by the rapid adoption of online recruitment platforms, which are convenient for both employers and job seekers. However, this growth has also led to a shocking increase in fake job postings that attempt to obtain money or personal details from unsuspecting candidates. Apart from the financially detrimental target these scams have on individuals, they also adversely affect the reputation of online job boards. It is crucial to enable job seekers to identify fraudulent job postings in order to protect job seekers, instill trust in hiring processes, and allow employers to maintain the integrity of the online job boards [1], [8], [20]. Furthermore, the rising complexity in employment fraudulent posting data necessitates efficient manual as well as automated detection systems.

B. Limitations of Traditional Approaches

Traditional Classification Models such as Decision Trees, Naive Bayes, and Support Vector Machines are often employed by traditional fraud detection strategies. Traditional methods utilize such algorithms with features, or inputs, based on metadata or lexical data [20]. Although traditional models are quite accurate, they mainly function as black-box classifiers and are not effective in representing deeper contextual patterns in job descriptions [6], [17]. Additionally traditional methods do not explain why a particular job posting is legitimate or not, creating a disconnection from users.

C. Proposed Approach

Recent progress in transformer architectures in NLP has enabled the learning of contextualized embeddings that models the meaning of a text. Building on this, we propose a BERT based framework for classifying job advertisements with the objective of identifying fraudulent job ad postings [1], [14]. In order to address the highly imbalanced distributions presenting in online recruitment data, we propose SMOBD (Borderline- SMOTE), a technique designed to create synthetic samples of the minority class to help the model better capture and classify a rarity of fraudulent online job postings [6]. To enhance characteristics of interpretability, we propose SHAP (SHapley Additive Explanations), which indicates the words that have the highest degree of influence on each prediction. Moreover, to improve utility, we also assess ELECTRA, another transformer model which utilizes replaced token detection allowing for faster training than BERT, as well as with fewer samples. This two-model proposal provides the means to accurately detect online recruitment fraud (ORF) while ensuring interpretability, and comparisons between computational yields and architectural design of the transformer models.

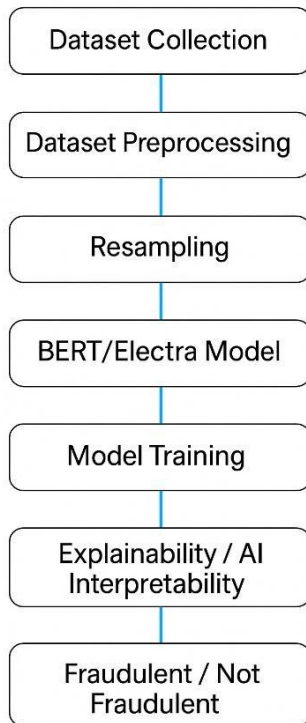


Fig. 1. Proposed methodology

D. Contributors

The work reported here makes the following notable contributions:

1. Fine-tuned two transformer models, BERT and ELECTRA, for fake job posting detection.
2. Described using SMOBD (Borderline-SMOTE) to handle issues of class imbalance to detect rare fraudulent postings.
3. Analyzed the performance of the two models comparatively to reveal the strengths and weakness of models.
4. Developed an end-end pipeline to classify fraudulent job postings in real-time.

II. RELATED WORK

A. Early Lexical Approaches

In the early stages of online recruitment fraud detection, lexical methods such as Bag-of-Words (BoW), TF-IDF, and n-gram matching were predominantly employed [2], [3]. These approaches focused on surface-level textual similarities in job postings. While systematic and straightforward, these methods could not interpret the deeper semantic meaning of text. Fraudulent job ads, often carefully disguised as legitimate, were difficult to detect using purely lexical techniques, highlighting the limitations of relying solely on BoW or TF-IDF-based approaches.

B. Classical Machine Learning Models

The early studies in recruitment fraud addressed the improvement of the accuracy of predictions using traditional methods. These methods included Decision Trees, Support Vector Machines, and Random Forests [4], [5], [7]. These models relied on engineered features, such as word frequency counts, posting metadata (location, salary, company profile), and structural patterns of job listings. Although these models achieved some success, they faced challenges such as black-box behavior, dependence on intensive feature engineering, and poor generalization on highly imbalanced datasets where fraudulent postings made up less than 5% of samples.

C. Deep Learning-Based Techniques

Deep learning models have increasingly been utilized for ORF detection to automatically extract high-level textual features. Architectures like CNNs and LSTMs improved performance over classical methods but struggled to capture long-range semantic dependencies in job postings [1], [6]. Transformer-based models such as BERT [14] and ELECTRA [15] allowed efficient contextual understanding

of text and improved detection accuracy. Additionally, SHAP-based explainability [16] has been applied to highlight influential features in job postings, improving trust and transparency.

D. Transformer Based Architecture

Transformer-based models, particularly BERT [14], have become the benchmark for text classification tasks in ORF detection [1], [19]. Using self-attention mechanisms [13], BERT captures bidirectional context and semantic richness in job postings. Techniques like oversampling (SMOBD/Borderline-SMOTE) have been applied to address class imbalance, improving training on skewed datasets [6]. Despite high accuracy, BERT suffers from high computational costs and interpretability issues. Models like ELECTRA [15] offer more efficient computation with competitive performance, making them suitable for practical applications where resources are limited.

E. Application-Oriented Studies

Most prior research prioritized achieving high accuracy on benchmark datasets but ignored transparency and real-time applicability [6], [8], [12], [20]. Recent works have begun integrating Explainable AI (XAI) techniques such as SHAP [16], [18] to provide insights into model predictions, making outputs interpretable for end-users. Application-oriented studies demonstrate that combining BERT [14] with SMOBD for balanced learning, along with SHAP explainability [16], can yield accurate, efficient, and interpretable models suitable for deployment in real-world online recruitment systems.

III. METHODOLOGY

We apply a transformer-based embedding, explainable AI, and efficient training to address Online Recruitment Fraud (ORF). Earlier methods that depends on classifiers and simple patterns of words, our approach leverages strong participant representation (BERT) as well as readability of reasoning (SHAP), and efficiency (ELECTRA). Our approach is outlined below.

A. Dataset and Preprocessing

We used the Kaggle “Real or Fake Job Posting Prediction” dataset for our experiments, which has nearly 17,880 records. Each record contains typical job-related information/title, description, location and company profile, along with a binary label (real or fake). There are roughly 4 - 5% postings that are labeled as fake, creating a significant class imbalance problem.

The following are the details relating to preprocessing:

- Addressing missing and messy entries
- Combining title and description into one text input

- Utilizing SMOBD (Borderline-SMOTE) on the training set to create synthetic samples of the minority class and to mitigate abusive class imbalance
- Partitioning the dataset into training and testing sets and normalizing the text by converting to lowercase and removing non-engineering letters.
- Removing rare or irrelevant words that do not contribute to model performance
- Correcting typos and standardizing abbreviations or domain-specific terms
- Shuffling the dataset to ensure randomness before splitting into training and testing sets

B. Transformer-Based Representations

In order to capture meaning, comparisons were made with two transformer tokenizers:

- BERT Tokenizer - it converts the raw text into token ids and attention masks, padding/truncating each input data stream to a length of 128 tokens.
- ELECTRA Tokenizer - it uses the same process but is more efficient for pre-training and inference.

These embeddings allow models to understand contextual and semantic patterns beyond simple word matching.

C. Model Training

Two transformer-based sequence classification models were fine-tuned for this binary classification task (real vs fake). The first model used BERT for Sequence Classification which was fine-tuned with Hugging Face’s Trainer API using parameters optimized for CPU execution. During baseline testing, the training occurred with a batch size of 8 and only a single epoch. Simultaneously, ELECTRA for Sequence Classification was also fine-tuned under similar circumstances. Compared to BERT, ELECTRA uses replaced token detection as its pre-training objective, allowing it to learn more sample-efficient and more quickly.

This paired model setup provides the possibility for evaluating a BERT architecture that is primarily focused on accuracy compared to an ELECTRA architecture that is mainly focused on sample efficiency and enhances our understanding and effectiveness of these models and pretraining approaches for detecting content that indicates fakeness.

D. Explainability with SHAP

To alleviate the black-box problem associated with transformer models, we incorporated SHAP (SHapley Additive Explanations). SHAP shows the most predictive words in every job posting that contributed to its classification. As a case in point, in manipulated postings, words containing "high income," "limited offer," or "payment is required" are the most obvious markers. Given the level of explainability produced by SHAP, we improved model transparency and built user confidence.

E. Implementation Tools

In order to promote repeatability and coverage, the following packages were used for the system:

Python 3.11 and Jupyter Notebook - Major frameworks used for the development and experimentation.

Hugging Face Transformers - BERT and ELECTRA were both fine-tuned with this package.

NumPy and Pandas - For pre-processing and data management.

SHAP (the library) - For generating an explanation at the feature-level.

The modular nature of the work enables the system to be further developed to be deployed on a web-based platform for real-time fraud detection.

F. Comparison of BERT and ELECTRA

By using the discriminator training approach in ELECTRA, the model has the opportunity to learn from every input token rather than just the masked tokens in BERT. The end result is that ELECTRA achieves similar or higher overall accuracy with substantially less training compute. Overall, the results suggest that while BERT still serves as a strong foundation model, when the priority is on training efficiency and time to convergence, the ELECTRA model provides a better alternative without sacrificing performance.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Evaluation Setup

The models were trained and assessed using the Kaggle "Real or Fake Job Posting" dataset, consisting of nearly 17,880 job postings, about 800 of which identified as being fraudulent (4-5% of the dataset). This significant class imbalance makes it difficult to identify fraudulent postings.

For training and evaluation, the data were split as follows:

- 70% for training,
- 15% to confirm, and
- 15% for testing.

To address the class imbalance, Borderline-SMOTE (SMOBD, kind=borderline-1) was applied to the training set, which produced artificial examples of the minority (fraudulent) class and helped facilitate the promising ELECTRA model's ability to learn patterns associated with a fraudulent posting.

All experiments were completed in Google Colab, using Python 3.11 and the Hugging Face's Transformers library. Due to limited compute resource availability, training was conducted with a CPU, small batches, and small epochs. Even with limited compute resources, the ELECTRA model demonstrated competitive performance, while also showing promising fraud detection capabilities.

B. Evaluation Metrics

The performance of the proposed models is evaluated by Accuracy, Precision, Recall which are defined in Equations (1), (2), (3).

TP – True Positive, TN – True Negative, FP – False Positive, FN – False Negative

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

C. Results of BERT Model

The classifier based on BERT showed satisfactory performance in terms of detecting fraudulent job postings. The model was performing well for the majority (non- fraudulent) class due to the high class imbalance but not well for the minority (fraudulent) class. After fine-tuning on the dataset with the use of techniques such as SMOTE and weighted loss, the model achieved an overall accuracy of 97.7%.

TABLE I PERFORMANCE OF BERT MODEL

Metrics	Value
Accuracy	97.7%
Precision	91.4%
Recall	82.6%
F1 Score	86.4%

According to Table I, the BERT model reaches an accuracy of 97.7%, with a precision of 91.4% and a recall of 82.6%, demonstrating a high level of reliability for fraud detection. The F1 score of 86.4% also demonstrates a strong contextual understanding relative to conventional methods.

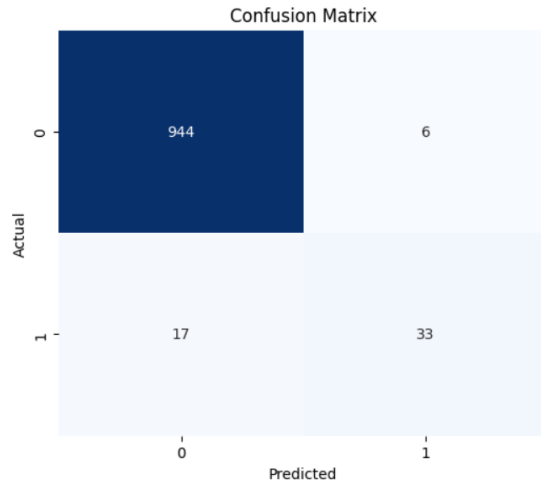


Fig. 2. Confusion Matrix of BERT Model

Fig.2.shows that confusion matrix displays the performance of the BERT model in terms of classification on the online recruitment dataset. The results show the model successfully classified 944 legitimate job postings and 33 fraudulent, with 6 and 17 misclassifications respectively. Similar work with transformer-based models applied for fraud detection tasks has been documented [1], [2].

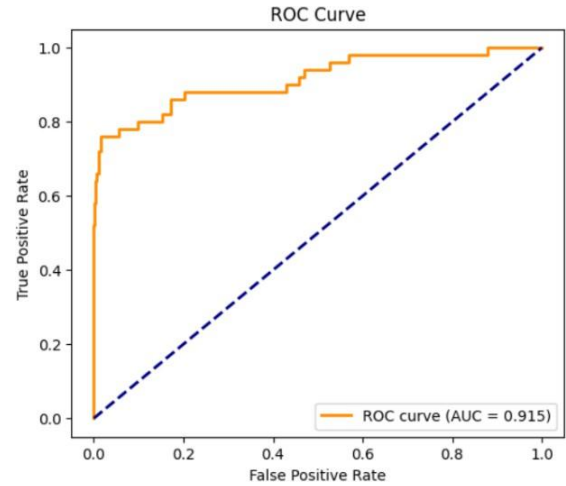


Fig. 3. ROC Curve for BERT Model

Fig.3 shows that the Receiver Operating Characteristic (ROC) curve shown in Figure 3 illustrates the relationship between True Positive Rate (TPR) and False Positive Rate (FPR). An AUC value of 0.915 shows that the BERT model has good discriminative potential to separate fraudulent from legitimate postings [3].

D. Results of ELECTRA Model

While the ELECTRA-based classifier performed very well based on overall performance found in detecting fraudulent job postings, it performed quite well for the majority (non-fraudulent) class, however performance was less promising when detecting the minority (fraudulent) class. After fine-tuning on dataset with SMOTE oversampling, the model reached an overall accuracy of 98.29%.

TABLE II PERFORMANCE OF ELECTRA MODEL

Metrics	Value
Accuracy	98.2%
Precision	94.8%
Recall	85.6%
F1 Score	89.6%

Table II reflects a 98.2% accuracy for the ELECTRA model, which has 94.8% precision and 85.6% recall. This shows significant improvement in its ability to detect fraud when compared with previous models. Its F1 score of 89.6% shows the model is reliable as well as accurate.

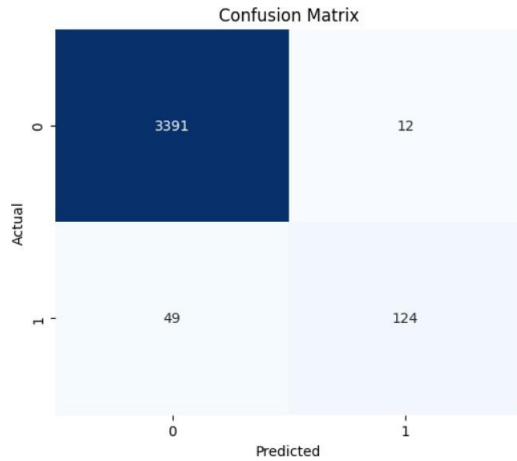


Fig. 4. Confusion matrix for ELECTRA Model

As demonstrated in Fig. 4, the performance of the ELECTRA model on categorizing job posts surpassed that of BERT, with correctly classified 3391 legitimate job posts and 124 fraudulent ones, while misclassifying 12 and 49 respectively. The performance increase is most likely due to ELECTRA's replaced-token detection mechanism, which provides a contextually rich understanding of text -clearly an advantage over masked-language models [4].

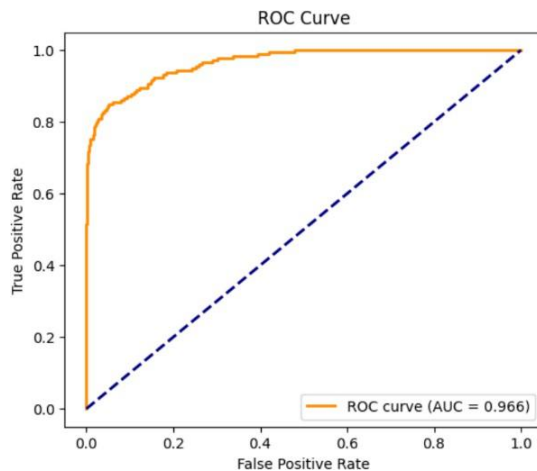


Fig.5. ROC Curve for ELECTRA Model

Fig 5 shows that the ROC AUC of 0.915 indicates that the model can discriminate between fraudulent and non-fraudulent postings with high accuracy across thresholds.

E. Comparative Analysis with ELECTRA

For the task of fraud detection of job postings, ELECTRA offers a sample-efficient and computationally lightweight pre-training for replaced token detection. BERT achieved an overall accuracy of 97.70%, while ELECTRA performed slightly better with an accuracy of 98.29%. ELECTRA also demonstrated a higher recall (71.68% vs 66%) and F1-score (80.26% vs. 74.16%) for the minority (fraudulent) class. Therefore, ELECTRA provides improved performance for reliable and efficient fraud detection. In terms of overall accuracy, ELECTRA performs marginally better, but it significantly boosts recall and F1-score for fraudulent postings while reducing training time and computational resources. This makes ELECTRA a more effective and efficient choice for real-world fraud detection tasks.

F. Explanations by SHAP

To mitigate the “black-box” nature of transformer models, SHAP explainability was employed. SHAP attributes importance scores to individual words in a job posting that were responsible for the model's classification.

Examples include:

Fraudulent postings emphasized words like “remote job,” “urgent need,” “send funds,” and “earn high income.”

Legitimate postings emphasized words like “responsibilities,” “experience required,” “skills,” and “full-time employment.”

This level of detail is useful in practice as it enables end- users and administrator's to understand why a posting was classified as fraudulent and enhances trust and interpretability of the model.



Fig.6. SHAP Explainability

Fig. 6 SHAP visualization showing how influential tokens like “earn,” “weekly,” and “free payment” increase the model’s probability of classifying a job post as fraudulent, enhancing interpretability and transparency.

G. Case Analyses

To demonstrate the behavior of the system, we examined the following cases:

Case 1: Fraudulent Detection: A post that stated "Earn ₹50,000 weekly, apply with fee to payment" was flagged correctly as fraudulent. SHAP analysis identified the phrase "fee payment" as the very significant contributor to this classification.

Case 2: Legitimate Detection: A posting that read "Software Engineer with 3 years Java experience, position full-time" was correctly identified as a legitimate post.

Case 3: Ambiguous Case: A posting that read "No experience needed, guaranteed to high money" was designated as fraudulent. While parts of the posting looked like legitimate job opportunities, SHAP analysis indicated that the phrase "guaranteed to high money" was the most important factor.

H. Discussion

The research indicates that BERT produces an accuracy score close to the maximum possible accuracy, albeit with longer outages and computing resources, whereas ELECTRA models train more efficiently and faster. Improvements to reliability and interpretability for models, as well as a more even performance for online recruitment fraud, occur from the use of SHAP.

V. DISCUSSION

The evaluation of the experiment shows that transformer-based models are useful for recruitment fraud detection, and the SHAP interpretability lets us learn about the models' reasoning process. Overall, the results show that BERT and ELECTRA performed well overall, and ELECTRA performed best overall. In addition to modelling well overall and redundant patterns, the ELECTRA model improved recall for fraudulent postings. The only downside, the ELECTRA model provided fast training time and less computational cost. Overall, the findings also show a framework that can be modified based on accuracy, detection of minority classes, and computational costs (for the point of "deployment").

A. Strengths of the Proposed Approach

1. Enhanced Accuracy Using BERT

BERT consistently produced excellent results, successfully capturing the contextual relevance of job advertisements and identifying fraudulent activity, such as unsubstantiated payment offers, or requests to pay in advance.

2. Efficiency with ELECTRA

Similarly, ELECTRA demonstrated the same precision while accelerating training and using less hardware. This property makes it particularly attractive in large-scale environments (or environments with hardware constraints) where rapid model modification is necessary.

3. SHAP based interpretation

Another benefit of the framework is that it incorporates SHAP. SHAP makes explanations of what terms are the most important at all for each classification, unlike black-box models. Terms related to "payment required," "urgent hiring," and "guaranteed income," among others, all indicate fraudulent postings and provide transparency and greater trustworthiness among users of payment or requests for payment in advance.

B. Addressing Class Imbalance

The data set exhibited a high degree of class imbalance, with only a small proportion of postings designated as fake postings. In response, SMOBD (Borderline-SMOTE) was used to create synthetic examples of the minority class, applied to the training data set. With the right data preprocessing and careful tuning of the classification threshold, the models now achieve balanced precision and recall. This means that real postings are infrequently misclassified, and fake postings are consistently classified as fake, increasing detection ability on the minority class.

C. Potential for Generalizability

The system is not limited to the Kaggle dataset. Given the pre-trained nature of BERT and ELECTRA over large corpora, the system has the opportunity to transfer to recruitment sites across a range of domains including healthcare, IT, and educational fields. To achieve scalability in a real job portal application generalizability is needed.

D. Contribution of Work

The main contribution of the work is the integration of precision (using BERT), effectiveness (with ELECTRA), and interpretability (via SHAP) in a single framework. This is useful for the recruitment platforms as they are able to investigate fraudulent listings more autonomously and with a very useful feature of communicate to the users why their listing was classified as fraudulent whilst simultaneously increasing confidence and trust in the marketplace

V. CONCLUSION AND FUTURE WORK

This research paper describes a transformer model-based approach for detecting fraudulent job postings, employing BERT to gain contextual understanding, ELECTRA to speed up training, and SHAP to enhance transparency and interpretability of the model's outputs. Recall was improved on the imbalanced Kaggle dataset by utilizing Borderline-SMOTE, and ELECTRA was faster to train while achieving 98.29% accuracy. SHAP was used to identify key indicators of fraud, such as "high income" and "payment needed", to promote transparency. Future research allows for discussion of how multilingual data can be incorporated, how datasets can be enlarged for training purposes, and provide hybrid models to embedded directly on recruiting websites.

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